From slant columns to vertical profiles

How do you get a profile out of a set of overlapping chords, and what does that step cost?

`Notebook companion: Unit-07-inversion.ipynb`

1 The question

Unit 6 delivered the primary product: species slant columns at each tangent altitude, with no atmospheric model in them. Most users want a vertical profile instead, and getting one requires undoing the overlap — every ray passes through its own tangent shell and every shell above it, so no measurement is of one altitude alone.

That step is not free, and the whole of the project's September reset came out of measuring what it costs. This unit sets out the two ways of doing it, measures what each one does to noise, and answers the question a reviewer will ask about the vertical-resolution claim.

2 Physical picture

Divide the atmosphere into spherical shells. A ray tangent at shell i passes through shell i and every shell above it, and through none below, so the measurement matrix is triangular. That triangularity is the opening: the topmost ray sees only the topmost shell and whatever lies above it, so it can be solved on its own, and once it is solved the next ray down has only one unknown left. Peeling downward this way is the oldest method in occultation work and it assumes nothing about the shape of the profile.

The alternative is to assume a shape. If the atmosphere is in diffusive equilibrium on a smooth temperature profile, the whole profile is described by four numbers, and those four can be fitted to every measurement at once. That is what `occ.py` does.

The two are not variants of one method. They sit at opposite ends of a trade. Peeling makes no shape assumption and therefore carries no shape bias, but it solves each shell from a difference between neighbouring rays, which amplifies noise and lets errors accumulate downward. Fitting pools all the information into four parameters, which suppresses noise dramatically, and pays for it with a bias whenever the assumed shape is wrong — which Unit 3 showed is precisely what a storm produces.

3 Derivation

For shells bounded by radii r_j, the two-sided path length of the ray tangent at r_i through shell j is

$$\Delta x_{ij} \;=\; 2\left[\sqrt{r_{j+1}^2 - r_i^2} \;-\; \sqrt{\max(r_j^2 - r_i^2,\, 0)}\right],$$

and the slant column of species s measured by that ray is

$$N_s(z_i) \;=\; \sum_{j \ge i} n_{s,j}\,\Delta x_{ij} \;+\; C_s(z_i),$$

where C_s is the contribution from above the topmost shell. That cap is not a detail. The topmost ray is constrained by nothing else, so the accuracy of the entire profile below it inherits whatever error is made in estimating the tail. `onion.py` estimates the cap scale height from the ratio of optical depths at the top two tangent heights rather than from a prior, so the boundary is measured rather than assumed.

Once shell densities are known, temperature follows without any further assumption. Integrating hydrostatic balance downward from a top boundary pressure gives $p(z)$, and $T = p/(nk)$. The error in the top boundary decays exponentially downward, so the temperature is trustworthy a couple of scale heights below the top of the retrieval and not at the top itself.

4 Worked numbers

7a. The geometry is the easy part

Ray	175	185	195	205	215	225
170 km	72.4	30.0	23.1	19.5	17.2	15.5
180 km	—	72.4	30.0	23.1	19.5	17.2
190 km	—	—	72.5	30.1	23.1	19.5
200 km	—	—	—	72.5	30.1	23.1

Path length of each ray through each shell, in units of the shell thickness of 10 km. Columns are labelled by shell centre.

A ray traverses 72 shell-thicknesses of its own tangent shell and 30 of the one above. That is Unit 5's limb enhancement, resolved shell by shell, and it is why the tangent shell dominates the measurement even though the ray passes through everything above it.

The condition number of the square system is 3.9. The geometry is not the ill-conditioned part of this problem; Unit 6's spectral conditioning and the noise amplification below are.

7b. What a held species does to the ones you can see

Running the peel on noiseless data, with the prior deliberately wrong by factors of two and four:

z	**Solved**	prior correct		prior ×2		prior ×4	
		O	N_2	O	N_2	O	N_2
175	O, N_2	0.3	10.5	50.5	−70.1	45.0	−100
205	O, N_2	1.7	6.7	19.3	−35.7	42.6	−97.7
225	O, N_2	2.0	5.7	9.8	−20.5	23.2	−67.4
235	O, N_2, O_2	2.2	4.6	2.4	4.4	2.7	3.8
275	O, N_2, O_2	2.4	3.8	3.0	1.6	4.8	−4.1
305	O, N_2, O_2	3.3	−0.3	6.4	−16.1	18.8	−80.2
325	O, N_2	8.0	−25.8	18.0	−100	18.1	−100

Percentage error in the retrieved shell density. The prior supplies starting values and fills any species the retrieval declines to solve.

Three things are visible and each matters.

With a correct prior the peel is accurate: oxygen to within 0.3 to 3.3 per cent over 175 to 305 km, with no assumption about the shape of the profile anywhere. That is the method working as advertised.

Where oxygen is solved along with the other two — above about 230 km, where the far-ultraviolet information is available — the answer barely moves when the prior is wrong by a factor of four: 2.2 per cent becomes 2.7. Where O_2 cannot be solved and is held at the prior, a factor-of-two error in that prior becomes a +50 per cent error in oxygen and a −70 per cent error in nitrogen. This is Unit 4's finding about unconstrained parameters appearing in the inversion: a species you cannot see does not politely drop out, it transfers its assumed error to the species you can.

And the top shell is the worst-determined one even with a perfect prior — 8.0 per cent in oxygen and −25.8 in nitrogen at 325 km— because it is constrained only by the estimated cap.

7c. What each retrieval does to noise

z	**input τ error**	**O out**	**N_2 out**	**gain O**	**gain N_2**
175 km	0.69 %	3.98 %	9.00 %	5.8	13.0
215 km	0.56 %	2.13 %	3.07 %	3.8	5.5
255 km	0.69 %	2.26 %	12.34 %	3.3	18.0
295 km	1.13 %	2.63 %	37.37 %	2.3	33.0
325 km	1.54 %	2.49 %	171.11 %	1.6	110.8

Onion peel, twelve Monte Carlo draws. "Gain" is the fractional error out divided by the fractional error in. Median gain for oxygen across the range: 4.0.

The peel multiplies the measurement error by about four for oxygen and by ten to a hundred for nitrogen, the latter rising steeply with altitude for the reason Unit 2 gave: above the crossover nitrogen is a shrinking minority of the absorbers and there is progressively less of it to see.

The parametric fit does the opposite:

z	input τ error	retrieved O spread	gain
200 km	0.55 %	0.61 %	1.11
250 km	0.69 %	0.46 %	0.68
300 km	1.34 %	0.27 %	0.20

Fitting four parameters to thirty-one sampled altitudes averages the noise down. The retrieved oxygen density at 300 km has a spread five times smaller than the measurement error that produced it.

`occ.ABEL_PENALTY` is a constant of 1.25 that the simulator applies to every reported error to account for this step, with a comment saying it should be replaced by a measured factor if the inversion were ever implemented explicitly. It has been, in `onion.py`, and the measurement above says two things. For the parametric fit the constant is conservative: the true factor is 1.11 at 200 km and falls to 0.20 at 300. For the peel it is far too small: the true factor is 1.6 to 7.8 for oxygen. The constant should be replaced by a per-retrieval, per-altitude factor, and doing so would slightly improve the precision figures we report above 250 km.

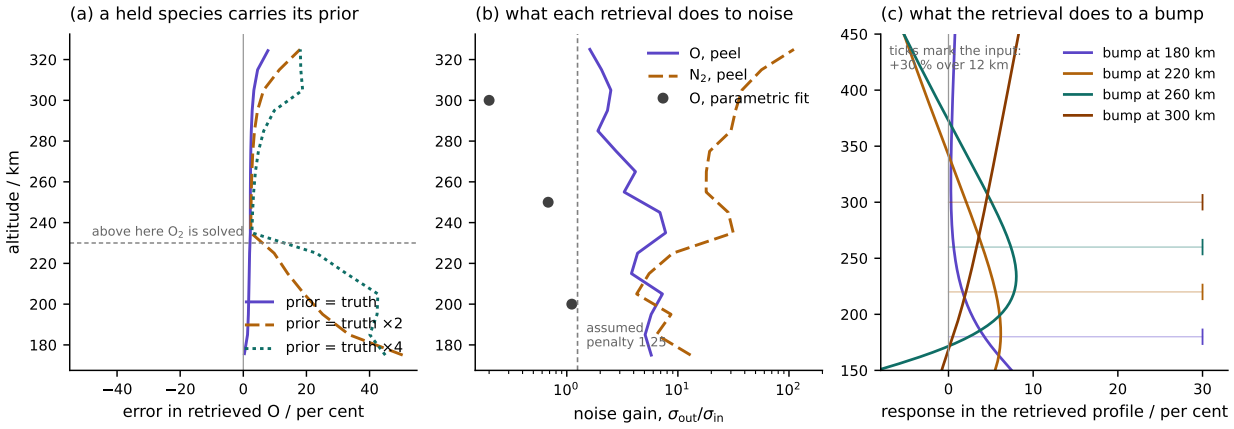


Figure 1: (a) Error in the retrieved oxygen density for three priors. Above 230 km, where O_2 is solved, the three curves coincide; below it they separate, because a held species carries its prior into the species that are solved. (b) Noise gain of each retrieval, with the assumed penalty of 1.25 marked. (c) The response of the retrieved profile to a narrow bump added to the truth, with ticks marking the input amplitude and altitude. Generated by sections B, C and D of the notebook.

7d. Averaging kernels

Add a bump of +30 per cent, 12 km wide, to the true oxygen profile; rerun the fit; look at the change in the retrieved profile. A retrieval that resolved 10 km structure would put the bump back where it came from.

Reported at	bump at 180	bump at 220	bump at 260	bump at 300
180 km	4.23 %	6.17 %	2.42 %	0.52 %
220 km	1.67 %	5.56 %	7.76 %	2.06 %
280 km	0.40 %	3.08 %	6.34 %	3.89 %
380 km	0.44 %	−1.90 %	−0.51 %	6.51 %
440 km	0.71 %	−4.80 %	−4.61 %	8.03 %

Change in the retrieved oxygen profile, in per cent, in response to a +30 per cent bump 12 km wide added to the truth at the stated altitude.

None of the four bumps is reproduced. A 30 per cent feature returns as a 6 to 8 per cent smear spread over 34 to 300 km, and in three of the four cases the peak of the response is nowhere near the bump: a bump at 260 km produces its largest response at 150 km, and one at 300 km produces its largest at 450.

WHY NOT we report sampling, not resolution

Detector rows binned to 24 µm subtend 4.7 mrad*, which is 10* km *at the tangent point, so the measurement is* sampled *every 10* km*. The retrieval is a different matter. It fits four parameters, so the retrieved profile lives in a four-dimensional space no matter how many altitudes are sampled — 31 sample points, four degrees of freedom. The averaging kernels above show what that means in practice: a 12* km *feature is returned at a fifth of its amplitude, smeared over tens to hundreds of kilometres, sometimes centred at the wrong altitude entirely. We therefore report a 10* km *sampling interval and an effective resolution of order a scale height, and we say which is which.*

5 What the computation shows

The two retrievals trade bias against variance, and the trade is large. The peel assumes no shape and multiplies noise by about four; the fit assumes a shape and divides noise by up to five. Neither is better. The right one depends on whether the atmosphere is doing something the assumed shape can follow, which is a question about the atmosphere and not about the algorithm.

A held species is not a neutral choice. Where O_2 is solved, a factor-of-four error in the prior changes the answer by half a per cent. Where it is held, a factor-of-two error changes oxygen by 50 per cent. Any retrieved profile has to be published with the prior that produced it.

The vertical resolution claim does not survive. Not because the sampling is coarse but because a four-parameter profile cannot represent 10 km structure, and the averaging kernels measure exactly how badly.

6 What we decided

The vertical-resolution row of the benchmark is reported as a 10 km sampling interval with an effective resolution of order a scale height, with the averaging kernels of section 7d as the evidence. This replaces the earlier claim and it is the correction the project identified on 5 September; this unit supplies the measurement behind it.

`occ.ABEL_PENALTY` is retired as a single constant. The measured factor is 1.11 at 200 km and 0.20 at 300 for the parametric fit, and 1.6 to 7.8 for the peel. Reported precision figures that used the constant are conservative above 250 km, and correcting them is a small improvement we can make honestly.

Every retrieved profile is published with the prior that produced it and with the list of species solved at each altitude. Section 7b is why: without both, a reader cannot tell whether a number is a measurement or an assumption.

The hybrid retrieval scoped as a Phase 1 deliverable — parametric in temperature, free in composition structure — is aimed exactly at the gap this unit measures. It is meant to keep the fit's noise suppression while removing the shape constraint on composition, which is the constraint that produces the storm-time bias.

7 What we do not claim

Not that the peel's accuracy with a correct prior is achievable. The 0.3 to 3.3 per cent in section 7b assumes the prior is truth, which in flight it never is. That row is there to show that the method is sound, not to predict performance.

Not that twelve Monte Carlo draws give a precise noise gain. They give a factor of about four for oxygen with a scatter that a larger ensemble would tighten. The conclusion — that the peel amplifies and the fit suppresses, by roughly an order of magnitude between them — is robust to the sample size; the individual entries are not.

Not that the averaging kernels are the full story. They were computed by perturbing the truth and refitting, which measures the response of this particular fit to this particular perturbation. A formal kernel would linearise the retrieval about its solution. The two agree in the respect that matters here, that the response is broad and small, but the numbers should not be quoted as a linear operator.

Not that hydrostatic temperature has been tested here. `onion.hydrostatic_temperature` is described in section 3 and used by the retrieval, and this unit does not measure its error. Unit 8 does.

8 Exercises

E7.1 Set the prior for O_2 alone to ten times truth, leaving O and N_2 correct, and confirm that the damage appears only below the altitude at which O_2 becomes solvable. State what that implies for how a prior must be reported alongside a retrieved profile.

E7.2 The peel's noise gain for oxygen rises towards the bottom of the range. Two mechanisms could explain it: accumulation of error from the shells above, or the loss of the aluminium channel below 227 km. Design an experiment that separates them, run it, and say which dominates.

E7.3 Section 7d used a 12 km bump. Repeat with widths of 30, 60 and 120 km and find the width at which the retrieval returns at least half the input amplitude at the right altitude. Report that width as the effective resolution and compare it with the local scale height from Unit 2.

References

[1] C. D. Rodgers, *Inverse Methods for Atmospheric Sounding: Theory and Practice*, World Scientific (2000). The standard treatment of averaging kernels and degrees of freedom for signal.

[2] R. K. Sewell et al., "Thermospheric composition and temperature from GOES-R/SUVI solar occultations," *Space Weather* **23**, e2024SW004234 (2025); arXiv:2508.10242.

[3] G. R. Smith and D. M. Hunten, "Study of planetary atmospheres by absorptive occultations," *Rev. Geophys.* **28**(2), 117–143 (1990), doi:10.1029/RG028i002p00117.

[4] E. M. B. Thiemann et al., "Neutral density profiles from PROBA2/LYRA solar occultations," *Space Weather* **15**, 1649 (2017); arXiv:1708.08833.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.